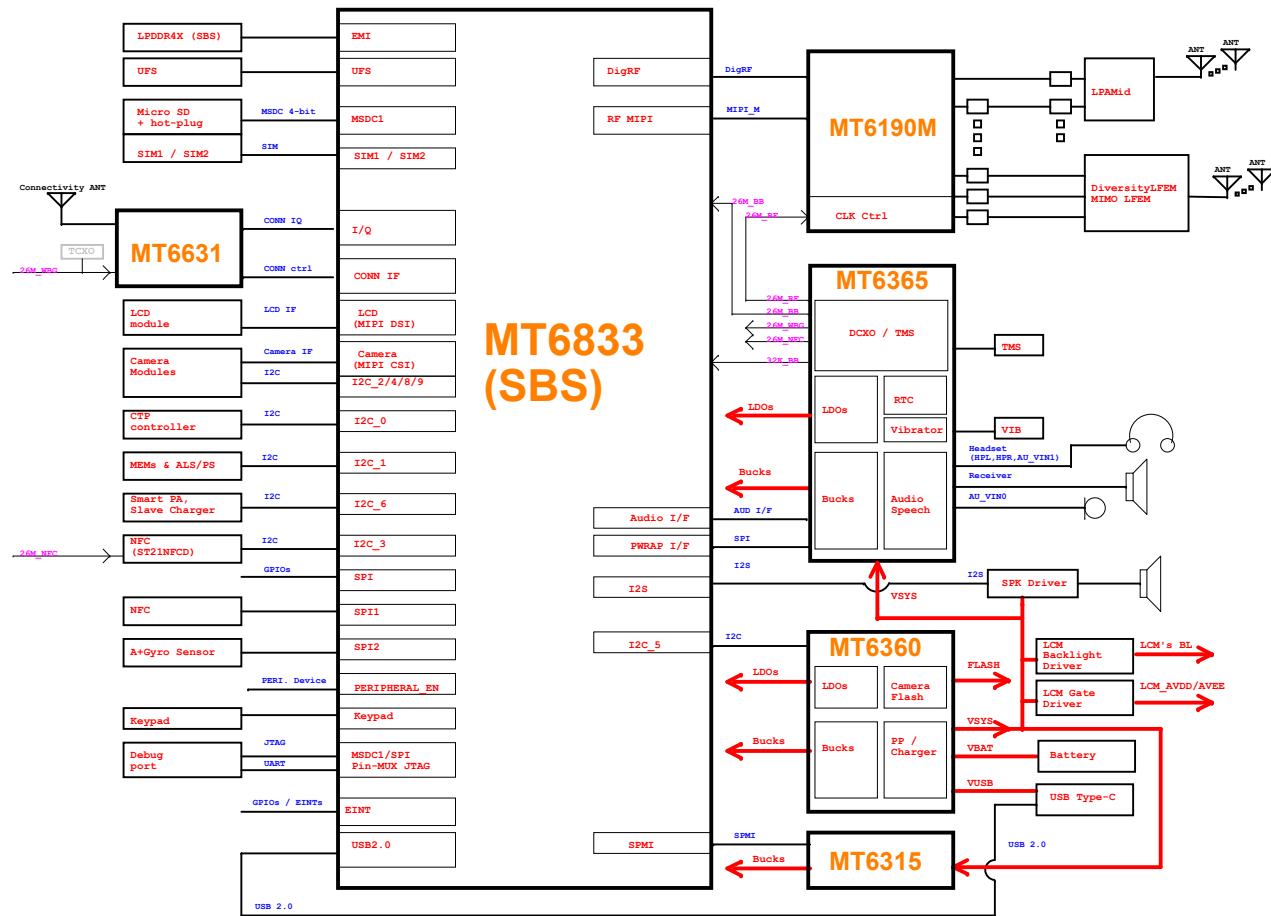
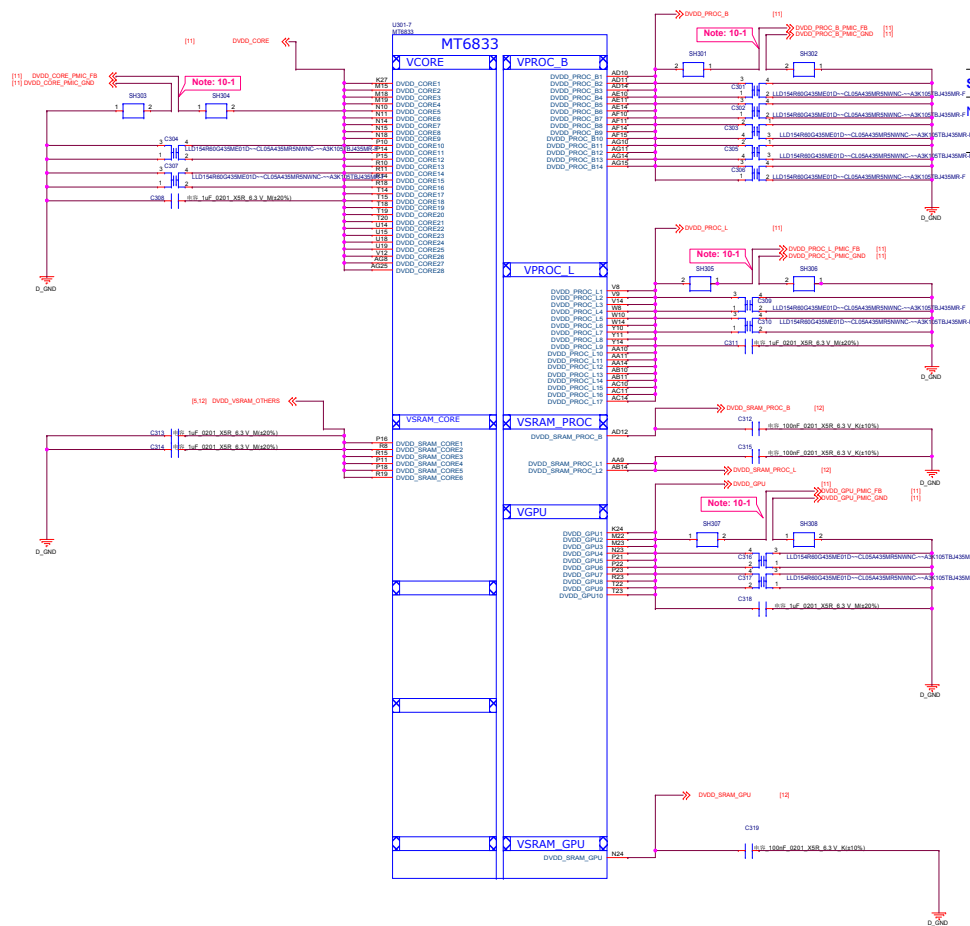




华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title Block Diagram		
Size D	Project <Doc>	Rev V1
Date: Tuesday, January 12, 2021	Sheet 1 of 48	

Date	Version	Page	Change notice
2020.06.xx	Pre-Released		
2020.07.xx	V01		
2020.08.14	V02	13_BB_1 29_POWER_EXT 20_POWER_MT6365_Buck 25_POWER_MT6315_VMD	Disconnect Dummy pins A1,A34,AP1,AP34 to GND Change C2923 cap from 6.3V to 10V [PDN] Change Caps of DVDD_GPU - C2039/C2038 to NC [PDN] Change Caps of DVDD_MODEM - C2502/C2534/C2523/C2522 to NC
2020.08.26	V03	File name	Modify the file name

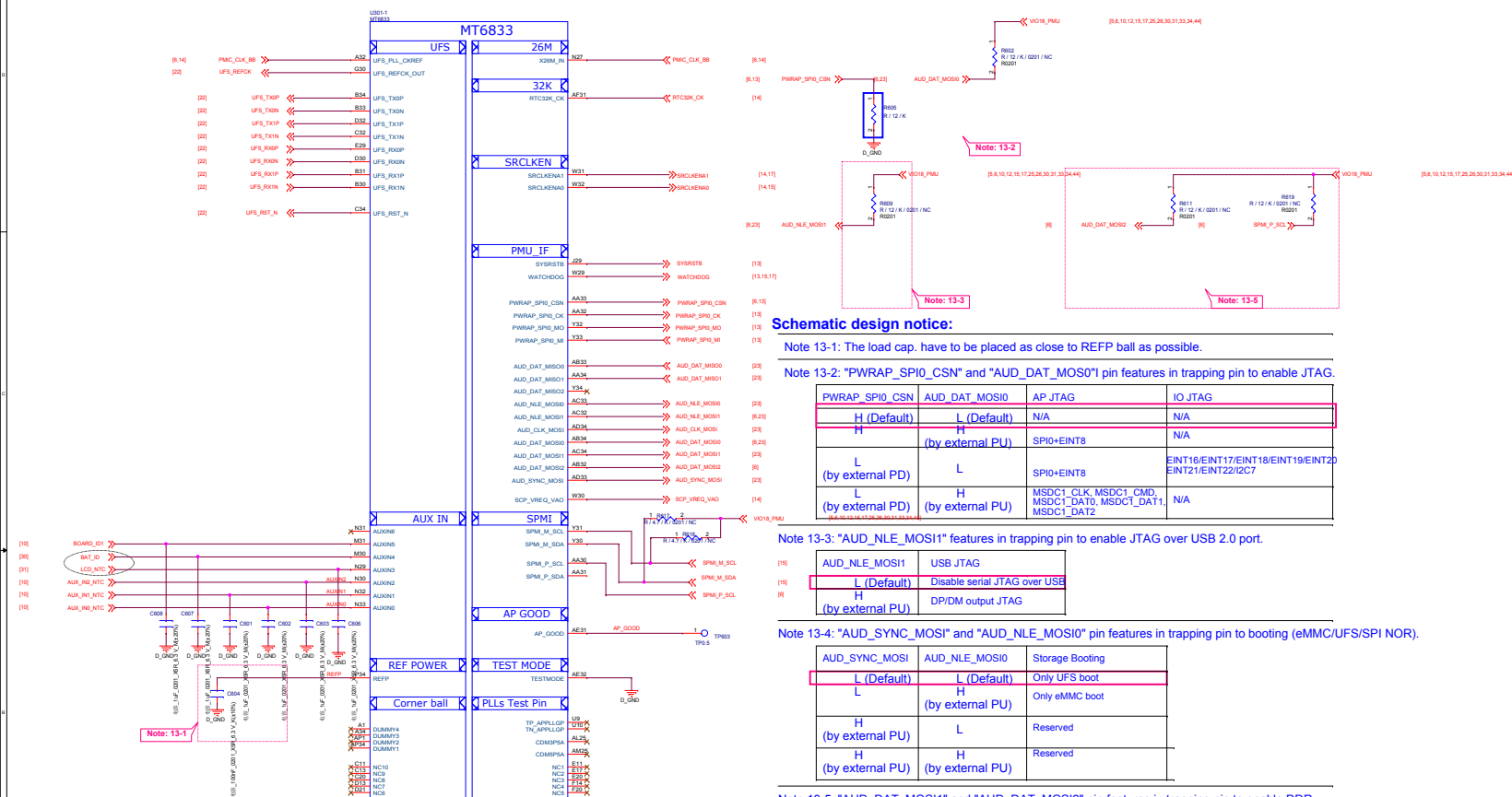
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title SM6350_Control		
Size D	Project <Doc>	Rev V1
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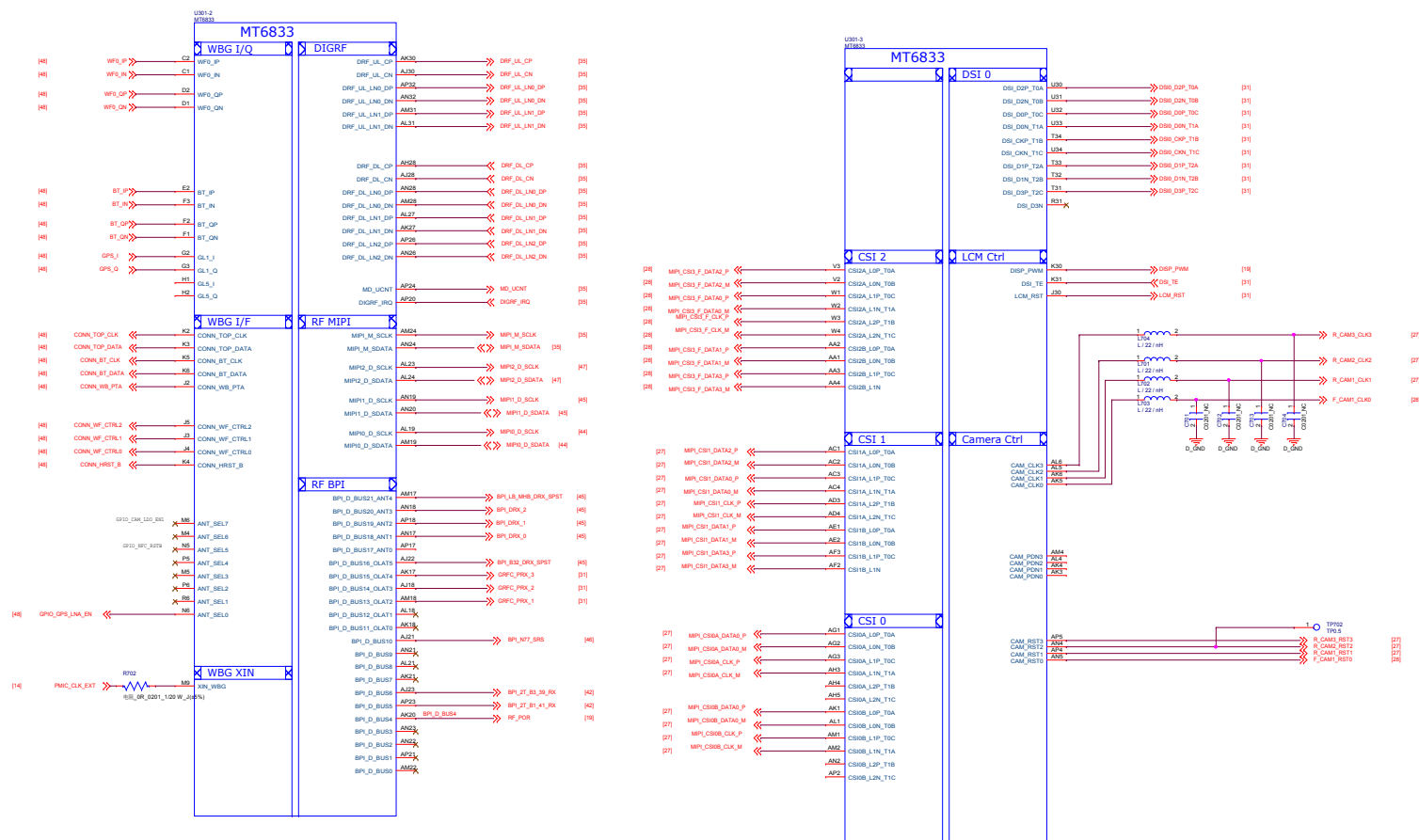
Schematic design notice of "10_BB_POWER_PDN" page.

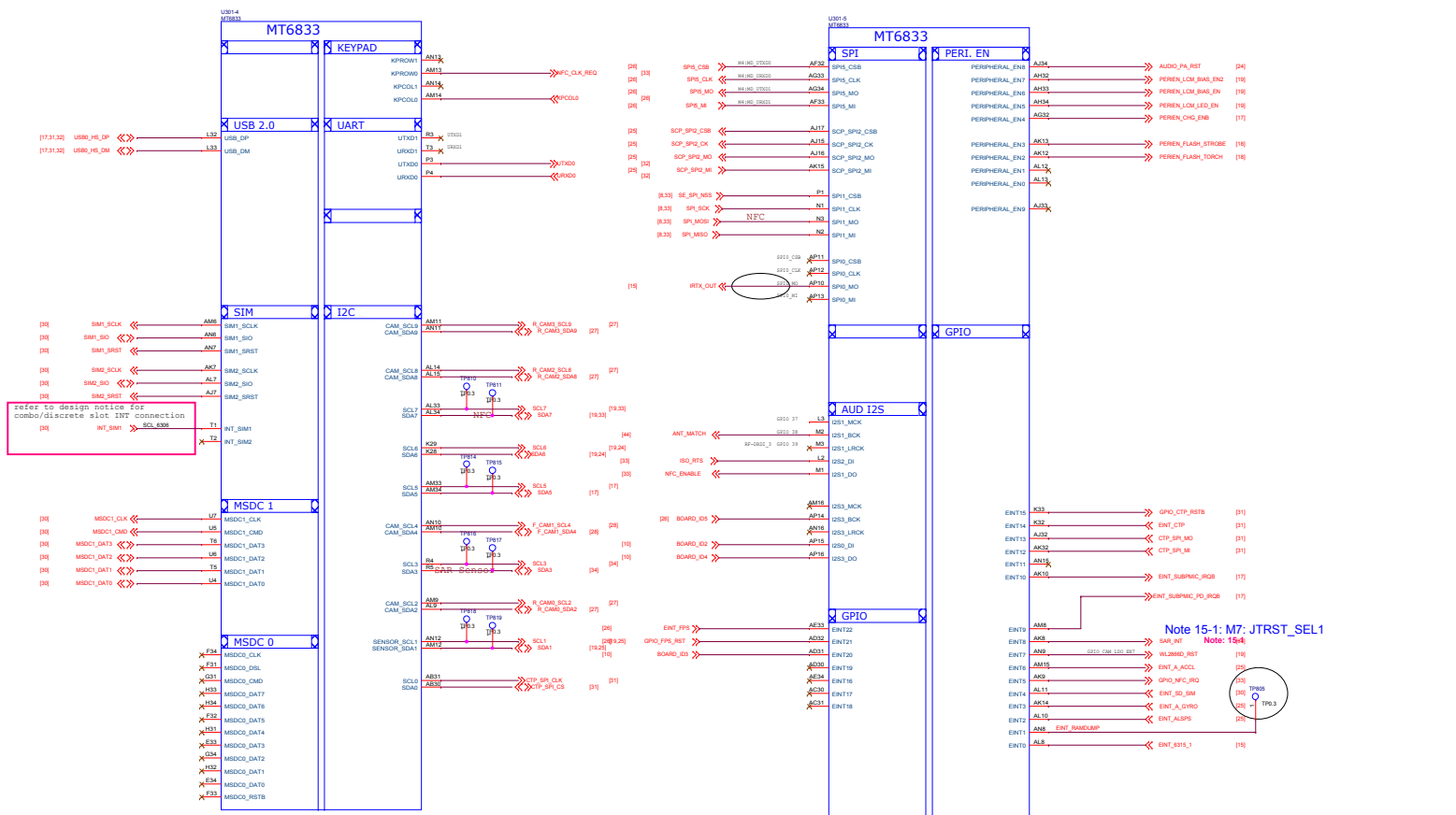
Note 10-1: Differential pair of PMIC remote sense must be close to BG's ball.
Remote sense trace with GND shielding to PMIC (Differential)

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 03.BB_POWER_PDN1			
Size D	Project 		

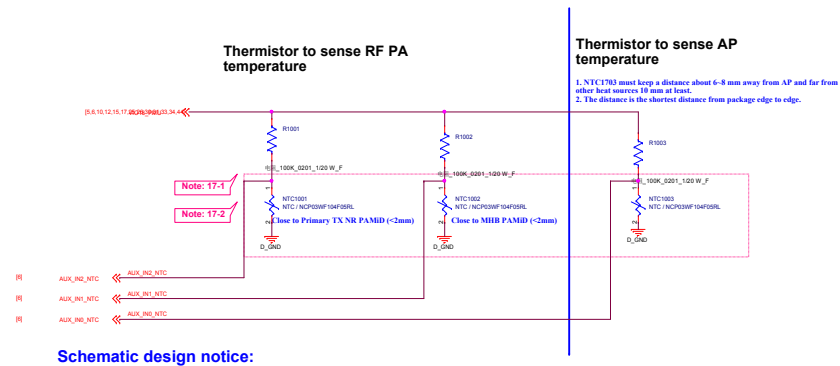


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title		
06.BB_1		
Size	Project	Rev
D	<Doc>	v1
Date:	Tuesday, January 12, 2021	Sheet 6 of 48





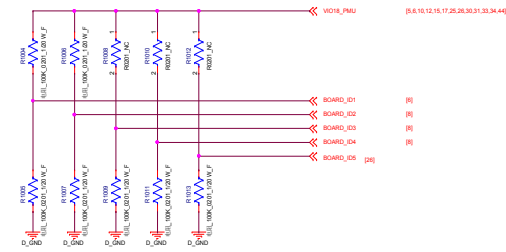
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		08.BB_3	
Size	Project	<Doc>	Rev
D			v1
Date:	Tuesday, January 12, 2021	Sheet 8 of 48	



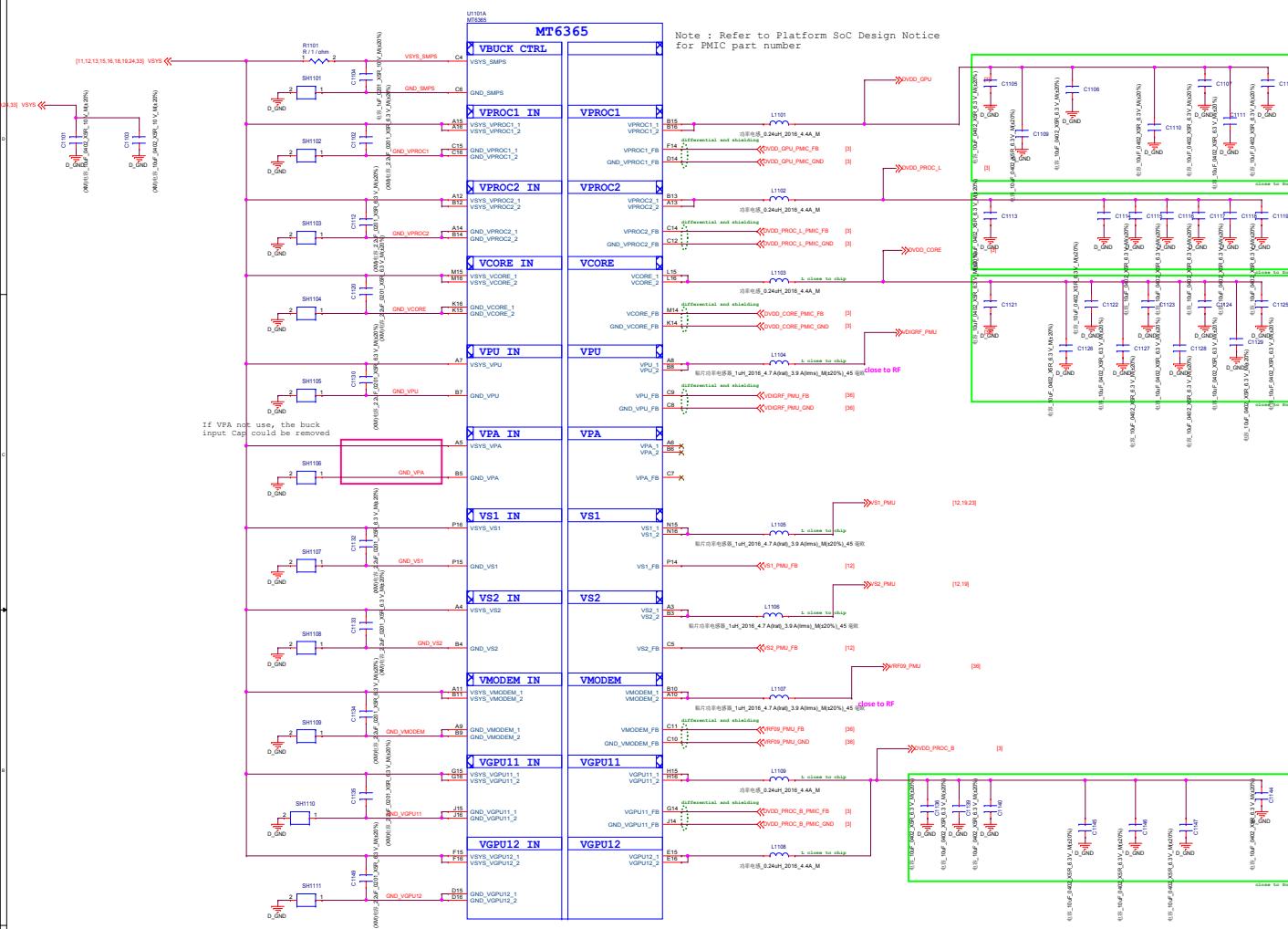
Schematic design notice:

Note 17-1: SW Default Configuration is as followings
AUX_IN0_NTC is for AP Temp.
AUX_IN1_NTC is for MHB PAMID Temp.
AUX_IN2_NTC is for NR PAMID Temp.

Note 17-2: If your design or placement is different with SW Default Config.
Please refer to "MTXXXX Thermal User Manuel.docx" and modify SW setting.



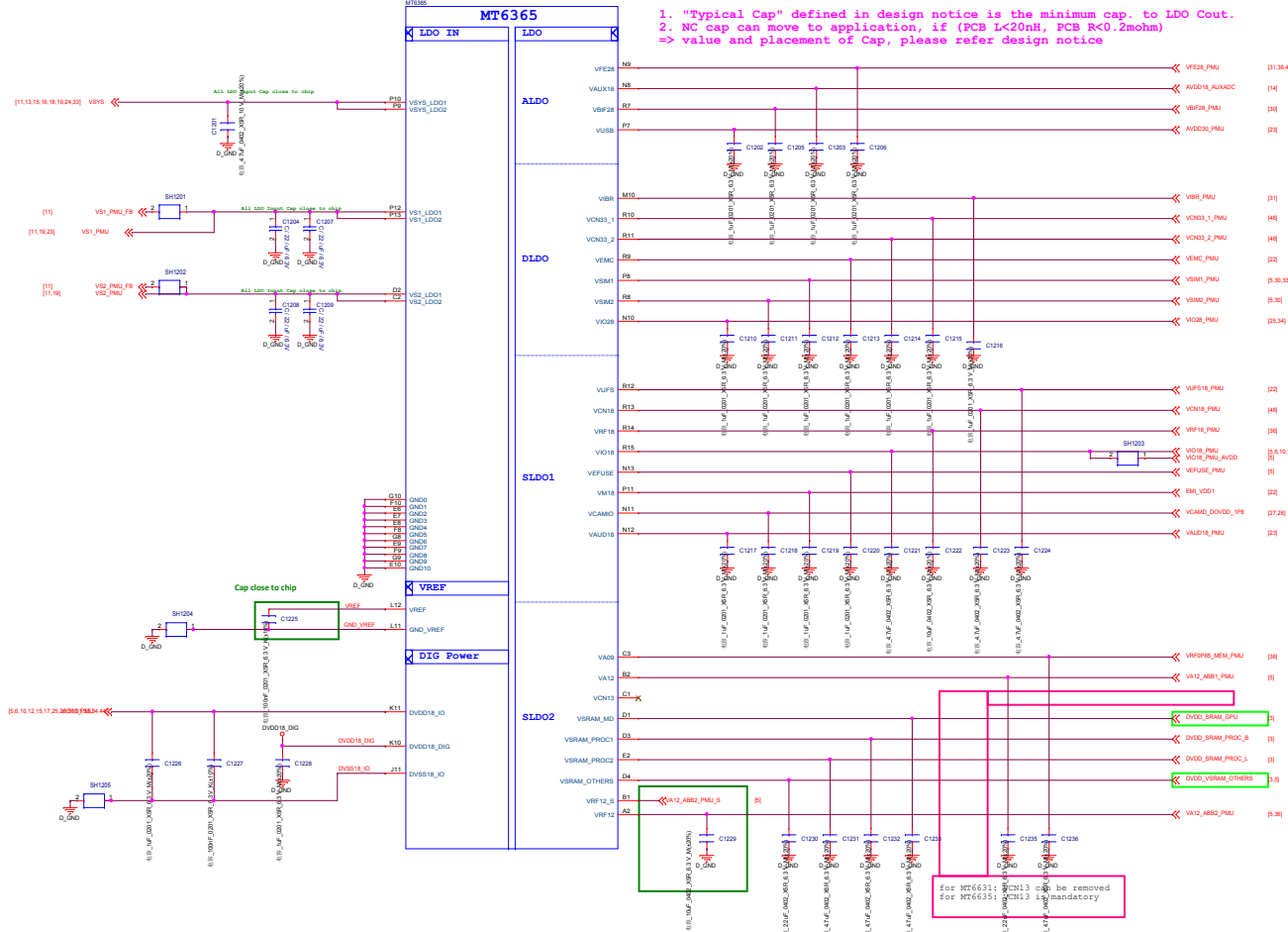
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 10.BB_AUXADC_Thermal/BOARD_ID			
Size D	Project <Doc>		Rev v1
Date: Tuesday, January 12, 2021	Sheet 10 of 48		



Circuit Type	Buck Name (Application name)	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)
Buck	VPROC1 (DVDD_GPU)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800
	VPROC2 (DVDD_PROC_L)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800
	VGPU11+12 (DVDD_PROC_B)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800 *2
	VCORE (DVDD_CORE)	0.40 ~ 1.3 (6.25mV/step)	ON (0.75)	4800
	VMODEM (VRF09_PMU)	0.50 ~ 1.10 (6.25mV/step)	OFF	4800
	VPU (VDIGRF_PMU)	0.40 ~ 1.19 (12.5mV/step)	ON (0.7)	2400
	VS1 (VS1_PMU)	1.86 ~ 2.20 (12.5mV/step)	ON (2.0)	2200
	VS2 (VS2_PMU)	1.20 ~ 1.50 (12.5mV/step)	ON (1.35)	2500
	VPA (VPA_PMU)	0.50 ~ 3.40 (50mV/step)	OFF	1000

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Title		11.POWER_MT6365_Buck	
Size D	Project <Doc>	Rev V1	
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Note : Refer to Platform SoC Design Notice
for PMIC part number



Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT,MAX} (mA)	Expected use
ALDO	VFE28	2.80	OFF (2.8)	200	RF/E
	VUSB	3.07	ON (3.07)	200	USB
	VAUX18	1.84	ON (1.84)	50	AUXADC
	VIO22	2.24	ON (2.24)	25	DCXO
	VBI28	2.80	OFF (2.8)	50	Battery Interface
VRTC	VRTC	2.80	ON (2.8)	2	RTC
VDIG	DVDD18_DIG	1.80	ON (1.8)	10	PMIC Digital

Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT,MAX} (mA)	Expected use
DLDO	VSIM1	1.7/1.8/1.86/2.76/3.0/3.1	OFF (1.86)	200	SIM
	VSIM2	1.7/1.8/1.86/2.76/3.0/3.1	OFF (1.86)	200	SIM
	VCN33_1	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VCN33_2	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VEMC	2.55/2.9/3.0/3.3	ON (3.0)	800	eMMC / UFS
	VIO28	2.8/2.9/3.0/3.1/3.2/3.3	OFF (2.8)	200	IO & Sensor
	VIBR	2.7/2.8/3.3	OFF (2.8)	200	Vibrator

Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT,MAX} (mA)	Expected use
SLD01	VEFUSE	1.80	OFF (1.8)	300	EFUSE
	VAUD18	1.80	ON (1.8)	300	Audio
	VCAMIO	1.80	OFF (1.8)	300	Camera IO
	VM18	1.80	ON (1.8)	300	DRAM
	VRF18	1.80	OFF (1.8)	450	RF
	VIO18	1.80	ON (1.8)	600	IO & Sensor
	VCN18	1.80	OFF (1.8)	1200	Connectivity
	VUFS	1.80	ON (1.86)	1200	eMMC / UFS
	VREF	1.40	ON (1.4)	10	DCXO
	VBBCK	1.24	ON (1.24)	10	DCXO

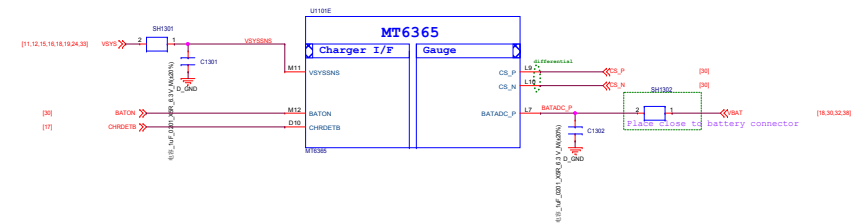
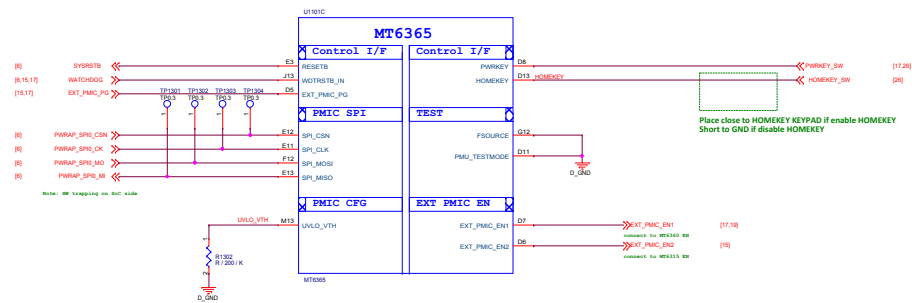
Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT,MAX} (mA)	Expected use
SLD02	VA09	0.85	ON (0.85)	300	DIGITAL SRAM (VRFOP85_MEM_PMI)
	VA12	1.2	ON (1.2)	300	AP Analog Module
	VCN13	1.3	OFF (1.3)	350	Connectivity
	VSAM_PROCI	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	CPUB SRAM
	VSAM_PROCI2	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	CPUL SRAM
	VSAM_PROCI3	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	OTHERS SRAM
	VSAM_PROCI4	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	GPU SRAM
	VSAM_PROCI5	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	GPU SRAM
	VRF12	1.2	ON (1.2)	800	AP Analog Module

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 12.POWER_MT6365_LDO

Size D Project <Doc> Rev V1

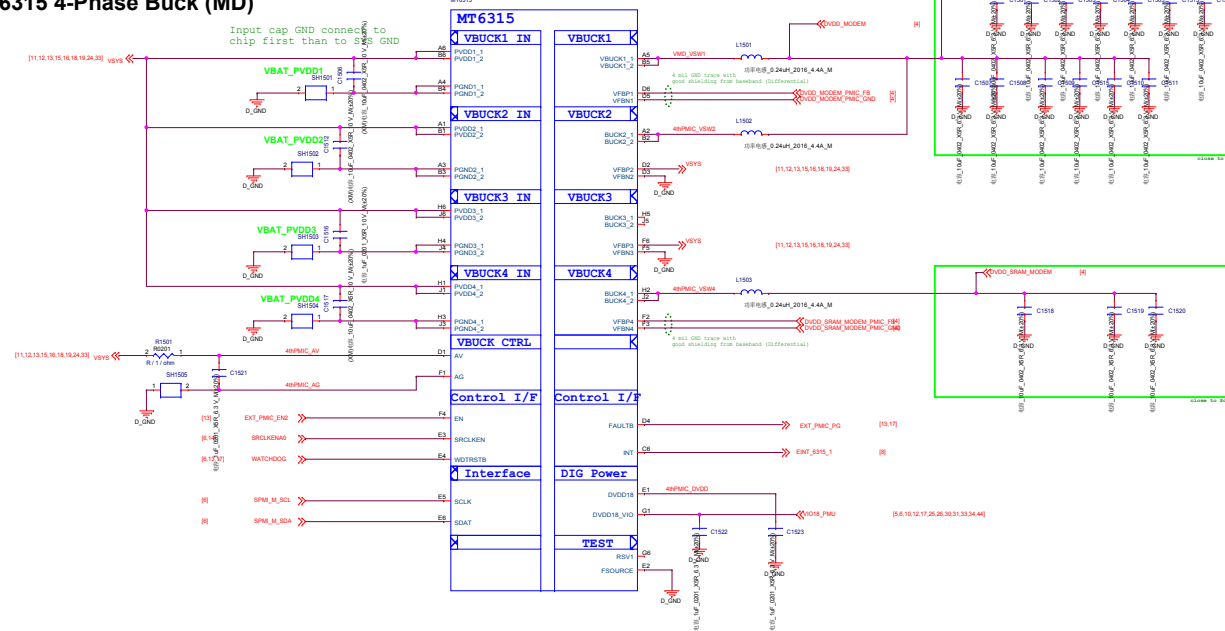
Date: Tuesday, January 12, 2021 Sheet 12 of 48



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		13.POWER_MT6365_General	
Size	Project	<Doc>	Rev
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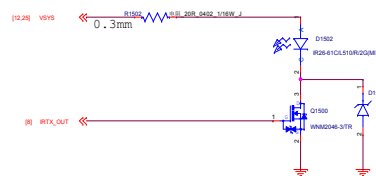
MT6315 4-Phase Buck (MD)

Note : Refer to Platform SoC Design Notice for PMIC part number

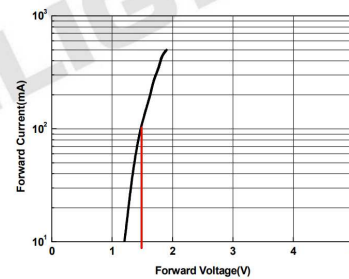


MT6315NP/B	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
BUCK1+2	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000*2	DVDD_MODEM
BUCK3	0.40 ~ 1.19375 (6.25mV/step)	ON (0.75)	5000	—
BUCK4	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000	DVDD_SRAM_MODEM

IR LED



Forward Current vs. Forward Voltage



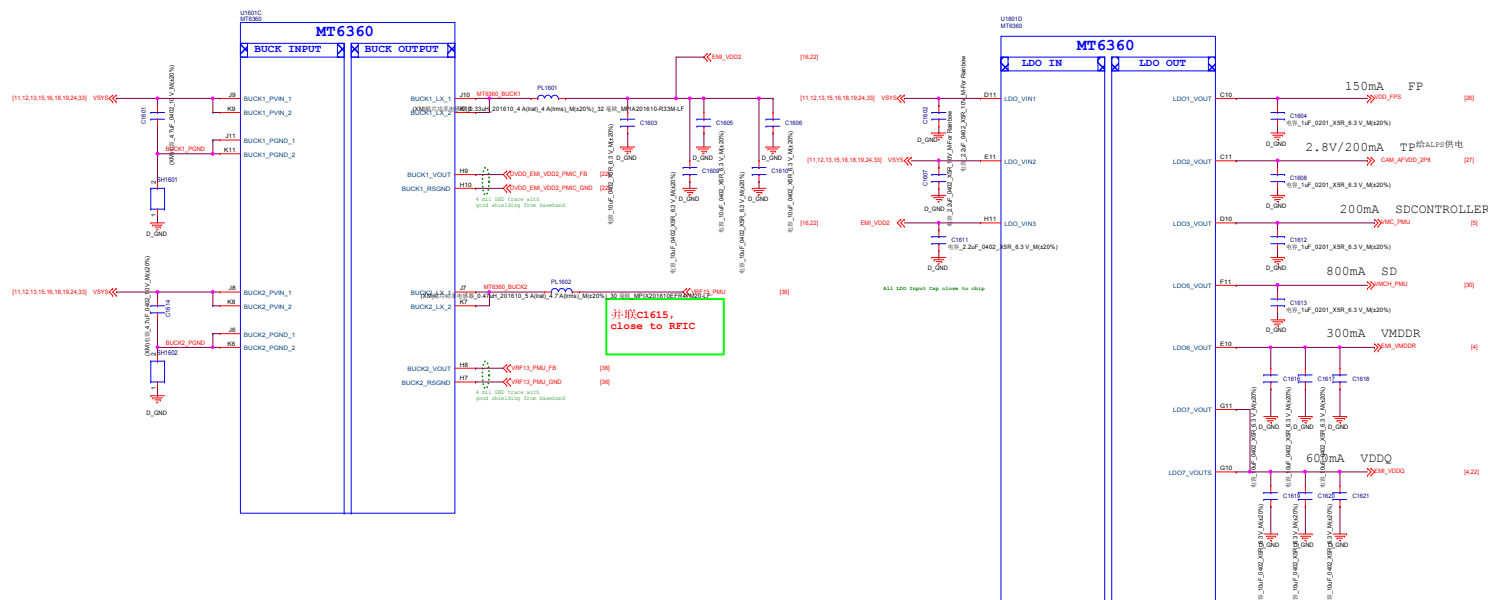
华勤通讯 Huaqin Telecom Technology Com., Ltd

Title 15.POWER_MT6315_VMD/IR_LED

Size D Project <Doc> Rev V1

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Note : Refer to Platform SoC Design Notice for PMIC part number



LDO1 VFP
LDO2 VTP
LDO3 VMC
LDO5 VMCH
LDO6 EMI_VMDDR
LDO7 EMI_VDDQ

Circuit Type	Name	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
Buck	BUCK1	0.3 ~ 1.3 (5mV/step)	ON (1.125)	3000	EMI_VDD2
	BUCK2	0.3 ~ 1.3 (5mV/step)	OFF (1.3)	3000	VRF13_PMU

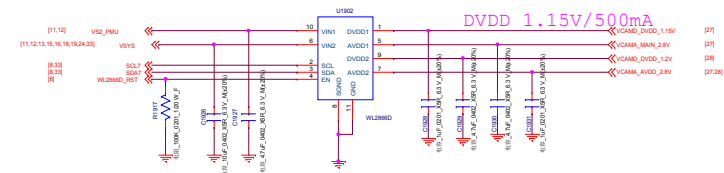
Circuit Type	Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
LDO	LDO1	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	150	Fingerprint
	LDO2	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	200	Touch Panel
	LDO3	1.8/2.9/2.8/3.0/3.3	OFF (3.0)	200	SD Card
	LDO5	2.9/3.0/3.3	OFF (3.0)	800	SD Card
	LDO6	0.75	ON (0.75)	300	EMI_VMDDR
	LDO7	0.6	ON (0.6)	600	EMI_VDDQ

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 16.POWER_MT6360_BUCK_LDO*		
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[illegible]

Note 28-1: For better ESD & surge performance we need choose suitable device for system protection. Please refer to the latest version of [Surge device selection guide] provided by MTK.

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 18.POWER_MT6360_Charger			
Size D	Project <Doc>		Rev V1
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[illegible][illegible][illegible]

厂家	IPC ADDRESS
KL2866D	0a50H
KT5904	0a50H

厂家	INC ADDRESS
OC92131	0111110 binary (or 3Eh)
SMC109	0111110 binary (or 3Eh)

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Title 19.POWER_EXT			
Size D	Project <Doc>		Rev v1
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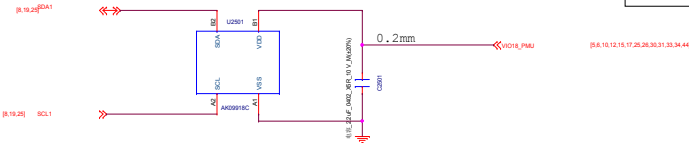
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title PM7250B_CHARGE/FG		
Size E	Project <Doc>	Rev v1
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华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title PM7250B_MISC/GPIO/GND		
Size D	Project <Doc>	Rev v1
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eCOMPASS

M-Sensor

厂家	I2C ADDRESS
AK8963C	0ch
AK8963C	2ch

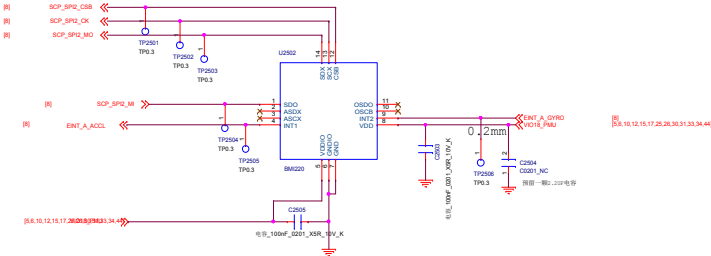


A+Gsensor

default SPI (if sensor hub support)

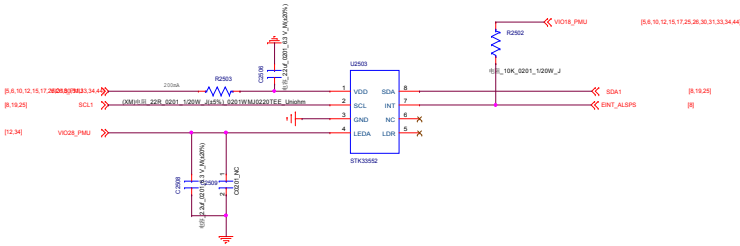
厂家	I2C ADDRESS
AK8963C	
AK8963C	

Gyro+A sensor



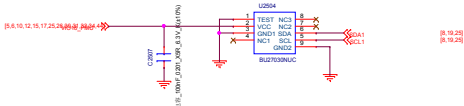
ALS+PS

厂家	I2C ADDRESS
AK8963C	0ch
AK8963C	2ch



后光感 BU27030NUC

厂家	I2C ADDRESS
AK8963C	0ch
AK8963C	2ch

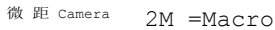
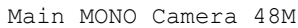


华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 25.SENSOR

Size D Project <Doc> Rev v1

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华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 27.REAR_CAM			
Size D	Project <Doc>		Rev v1
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Title		Sub_Charger_SMB1396	
Size D	Project <Doc>		Rev v1
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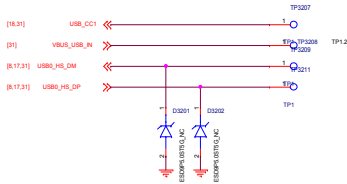
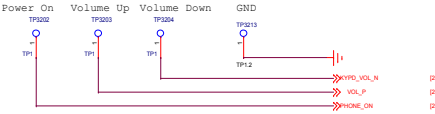
VBAT TEST POINT



VBAT

KEYPAD

KEY BTB

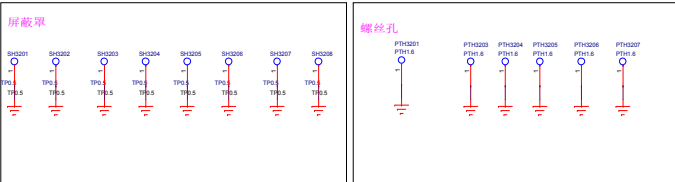


VCHG

UART TP

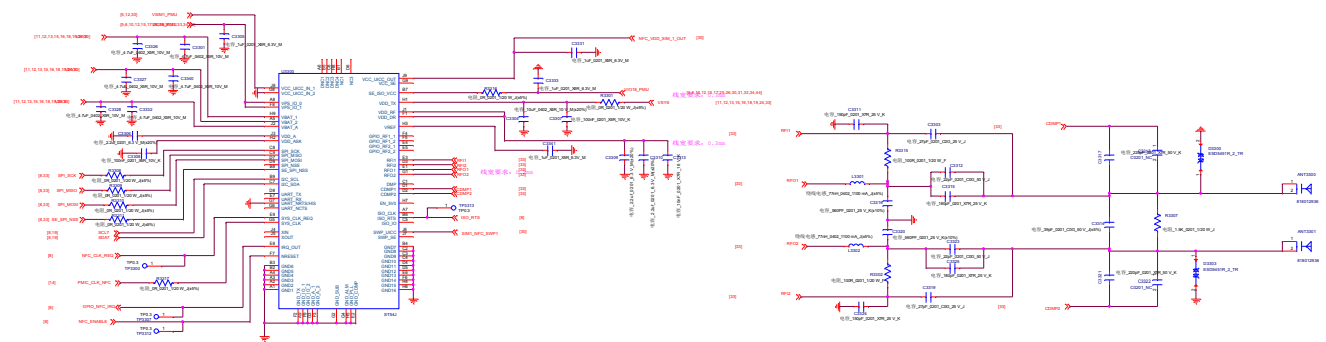


JTAG



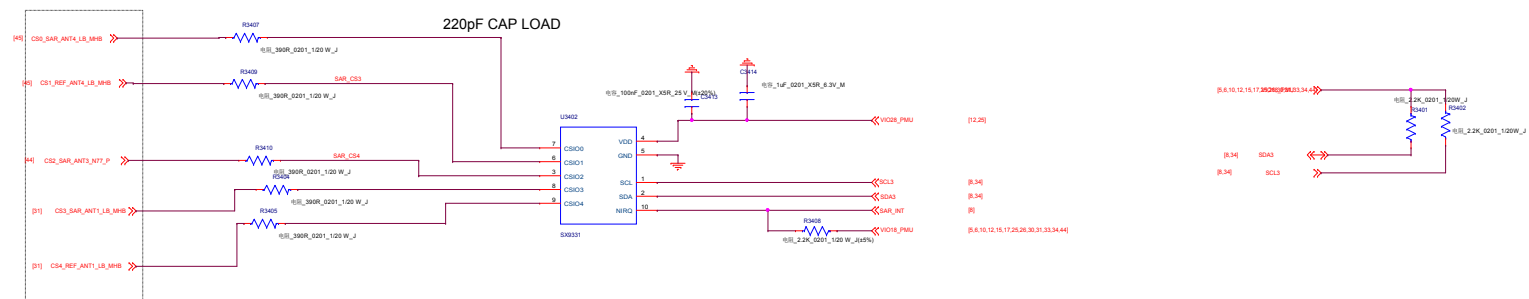
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 32.TEST_POINT/SHIELD		
Size D	Project <Doc>	Rev v1
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RF_NFC_not support



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		RF_NFC	
Size	Project	<Doc>	Rev
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Date: Tuesday, January 12, 2021		Sheet 33 of 48	

SAR_sensor Not support



Design notice:

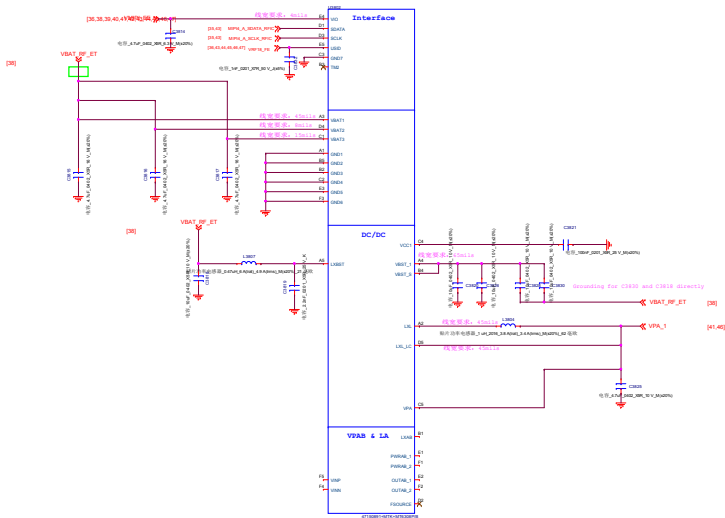
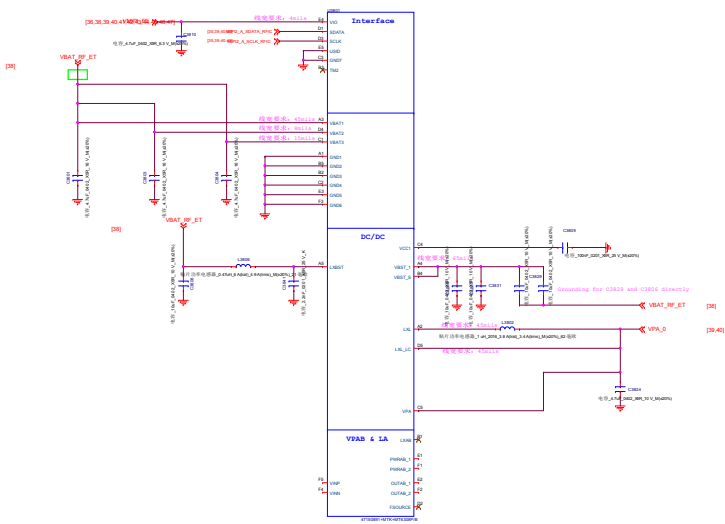
- 1) CS0_1 nets are differential pairs;
- 2) The width for each line, should be less than 0.1mm;
- 3) 0.2mm should be guaranteed between the ground and differential pairs;
- 4) And also, the distance between the differential line should be more than 0.2mm;
- 5) All the CSx nets should be protected by surrounding GND;

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title SAR_sensor			
Size D	Project <Doc>		Rev v1
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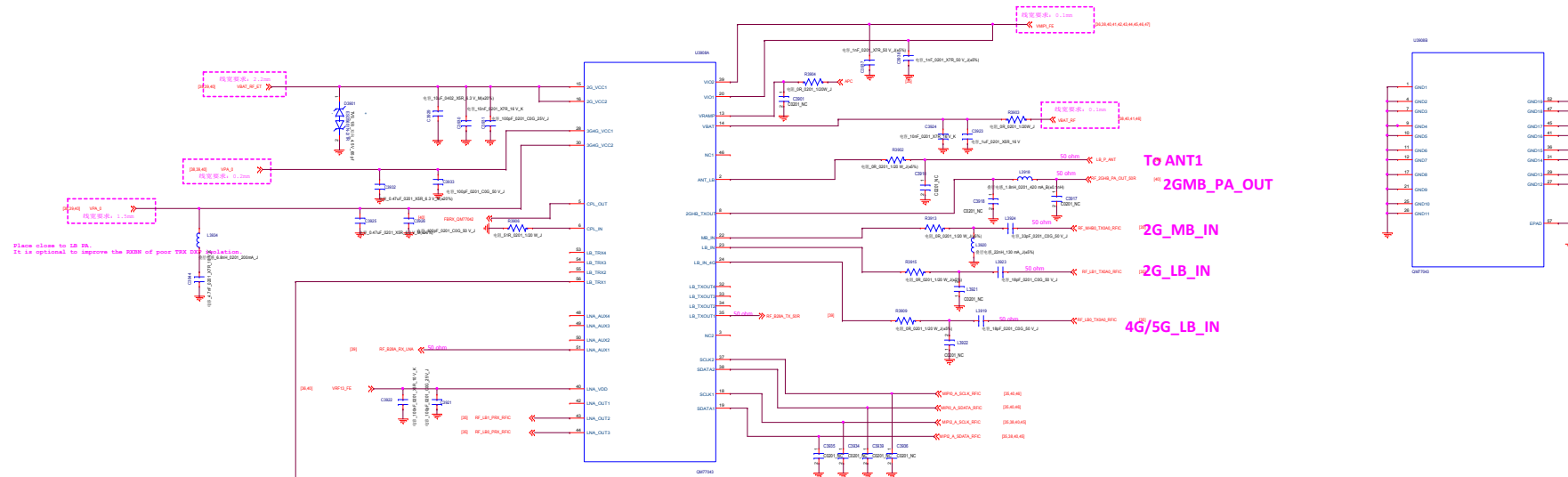
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title NC_FBRX			
Size D	Project <Doc>		Rev v1
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QM77042 and QM77048 for one APT

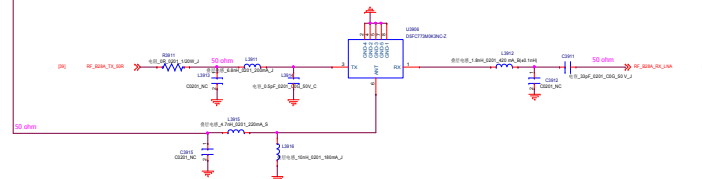
VC7643 and QM78200 for the other APT



2G_3G/4G/5G LB PA_QM77042



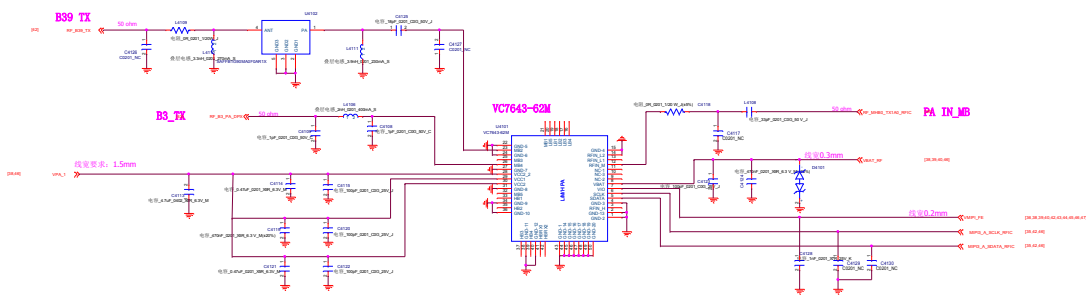
QM77042同QM77043 pin to pin



NTC温度检测
见P10的R1001 and NTC1001

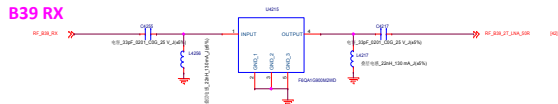
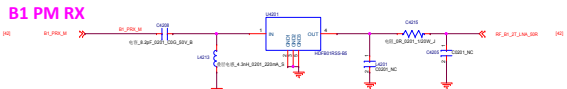
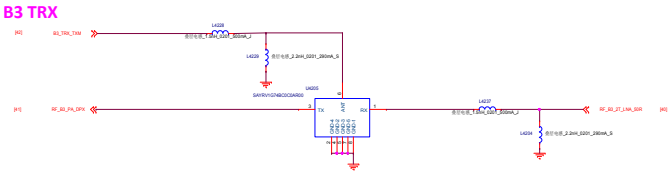
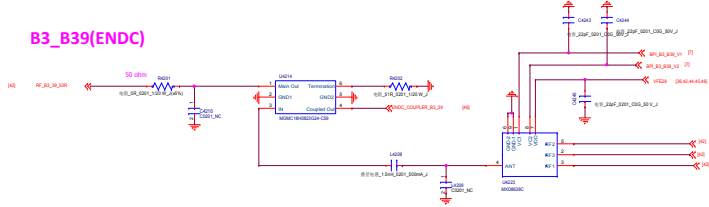
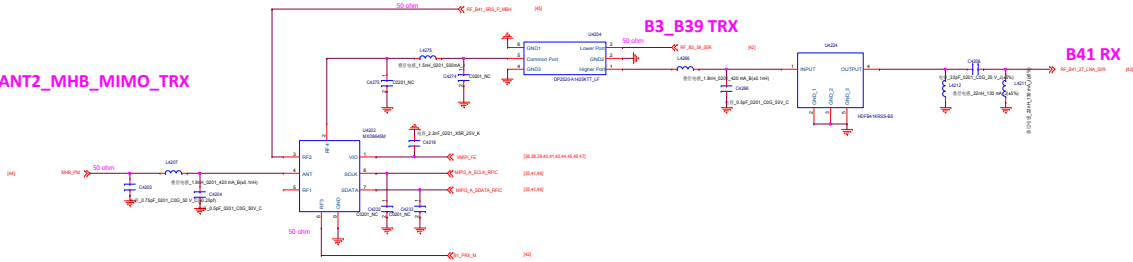
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title QM77042			
Size AQ	Project <Doc>	Rev v1	
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MMMB PA_MB ENDC



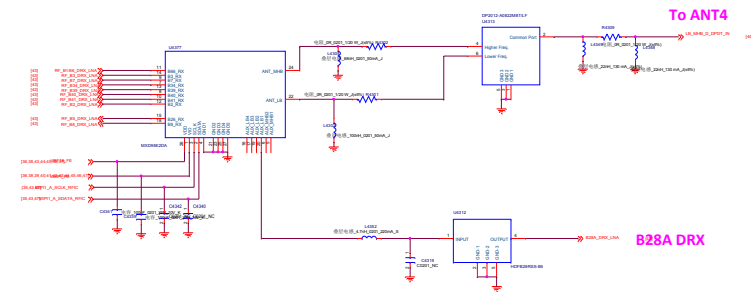
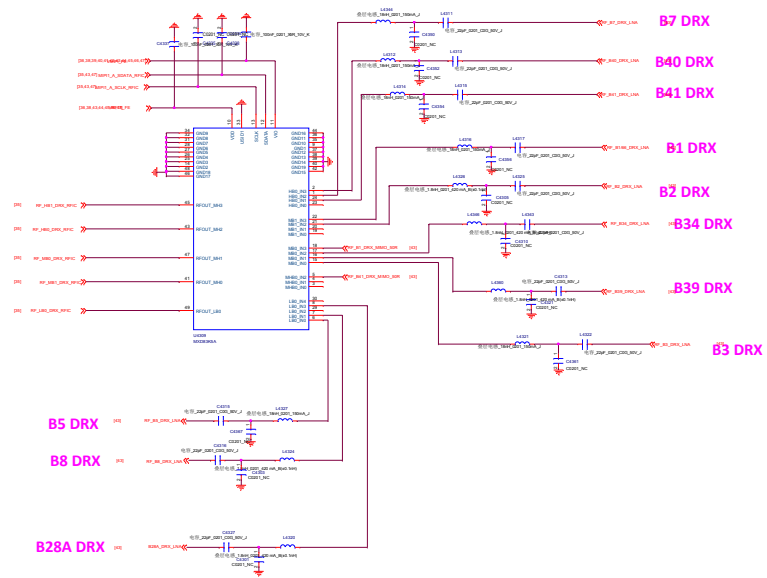
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		MMMB PA_MB ENDC	
Size	Project	<Doc>	Rev
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MMMB PA_VC7643_ENDC_MHB_TRX

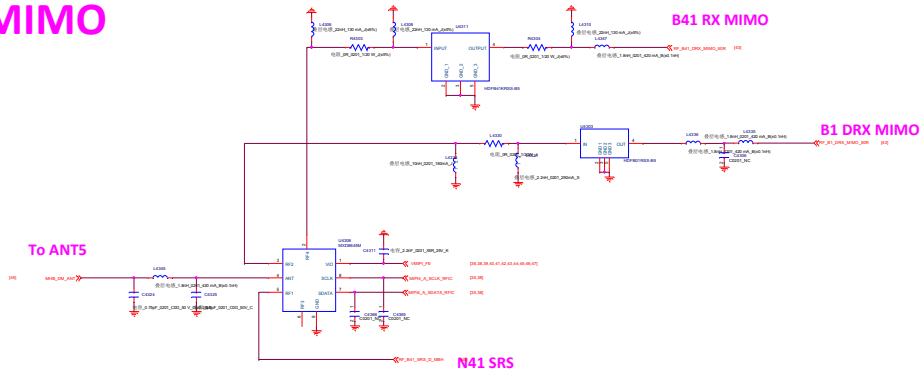


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title MMMB PA_VC7643_ENDC_MHB_TRX		
Size A0	Project <Doc>	Rev v1
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DRX_MXD98E2DA+MXD83K5A

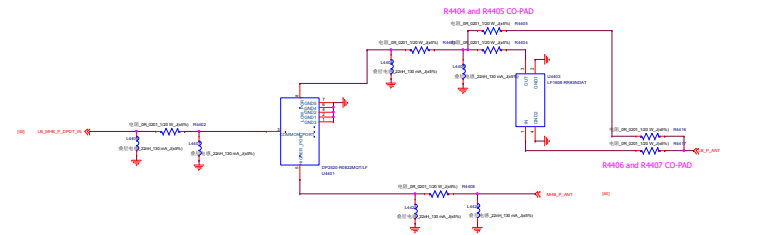
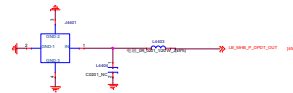
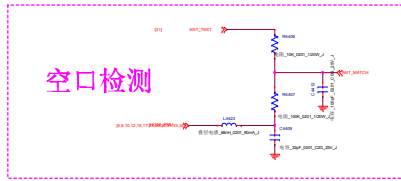


DRX_MIMO

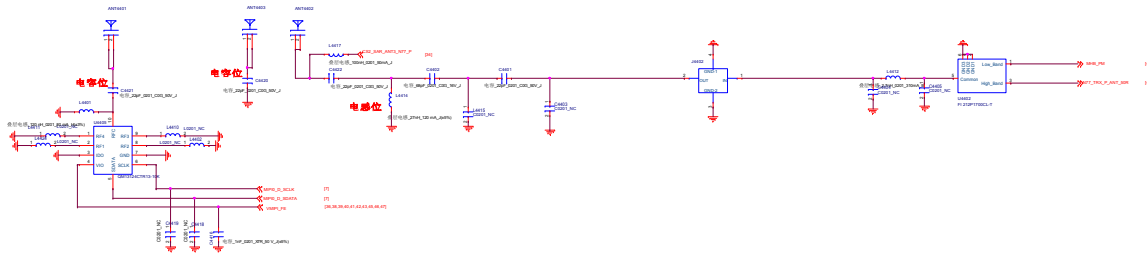


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title			
MXD98E2DA_MXD83K5A			
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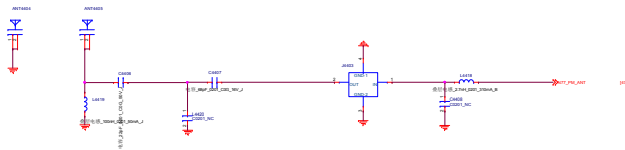
ANT1 (LB_MHB_P) to 小板



ANT2 (MHB_PM/N77_TRX)

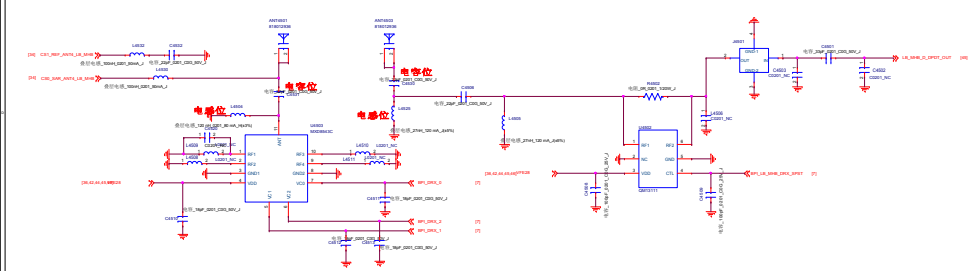


ANT3 (N77_PM)

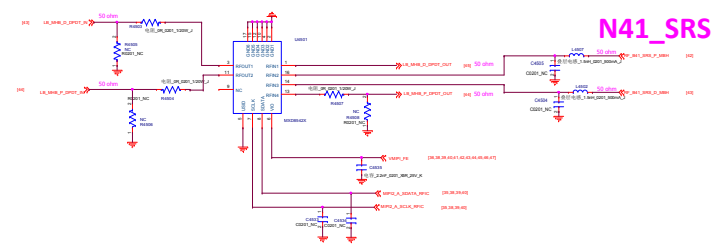


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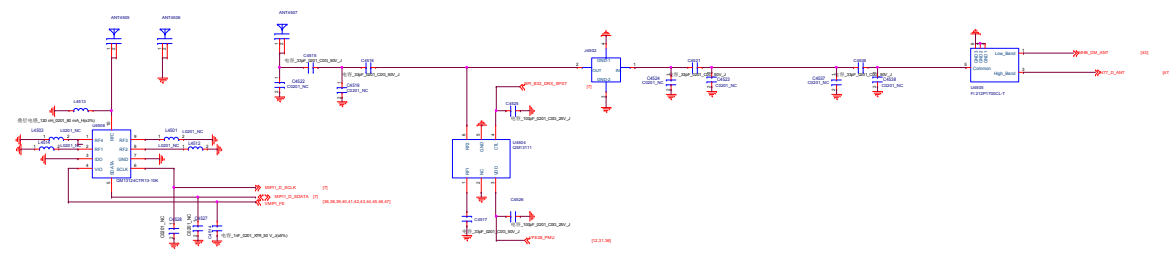
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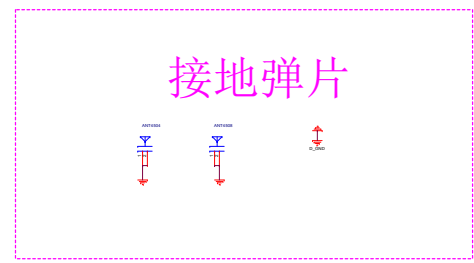
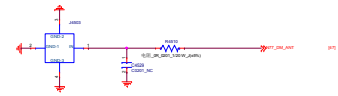
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ANT5 (MHB_DM/N77_D)

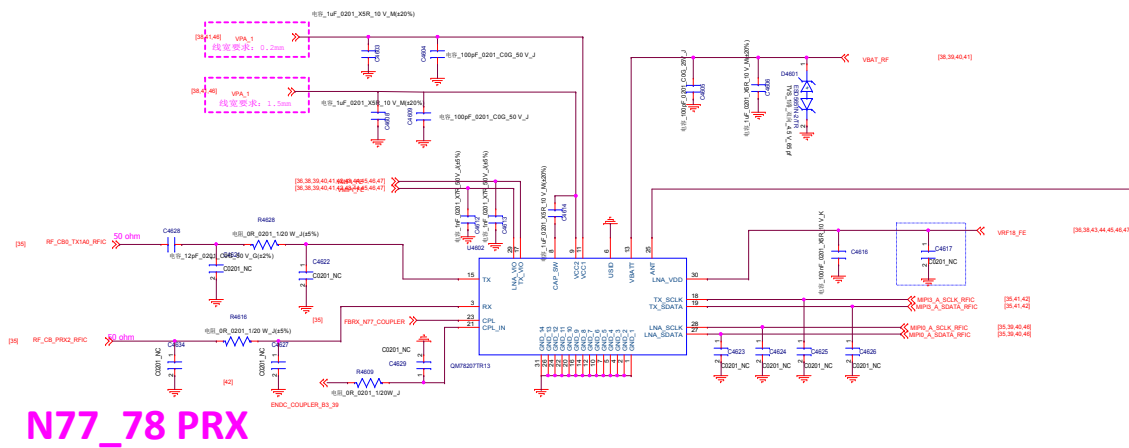


ANT6 (N77_DM) to 小板



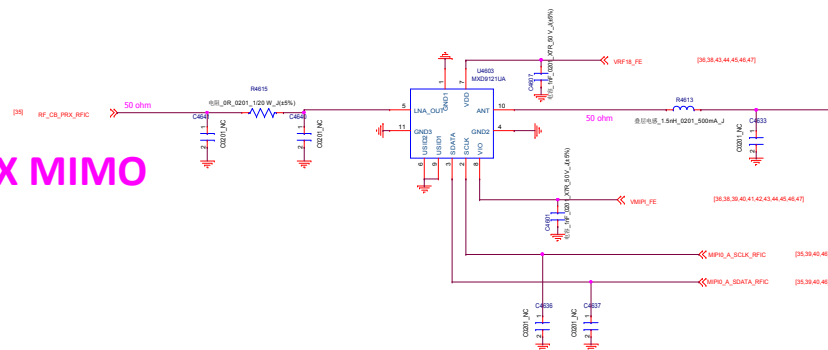
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Title ANT_D_DM			
Size E	Project	<Doc>	Rev V1
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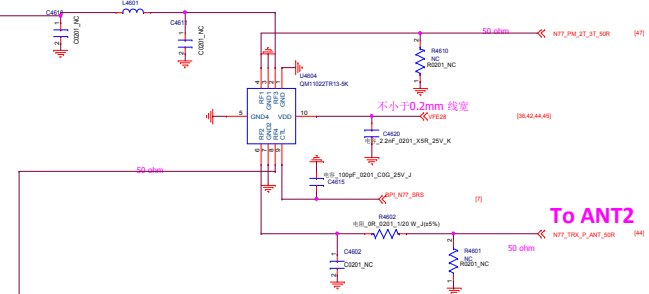
N77/78_MIMO_MXD9121UA

N77_78 PRX MIMO



NTC温度检测
见P10的R1002 and NTC1002

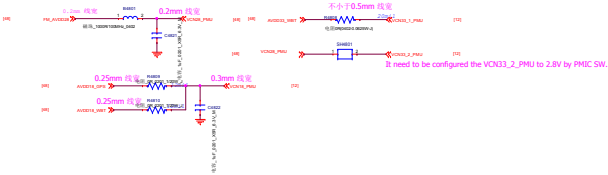
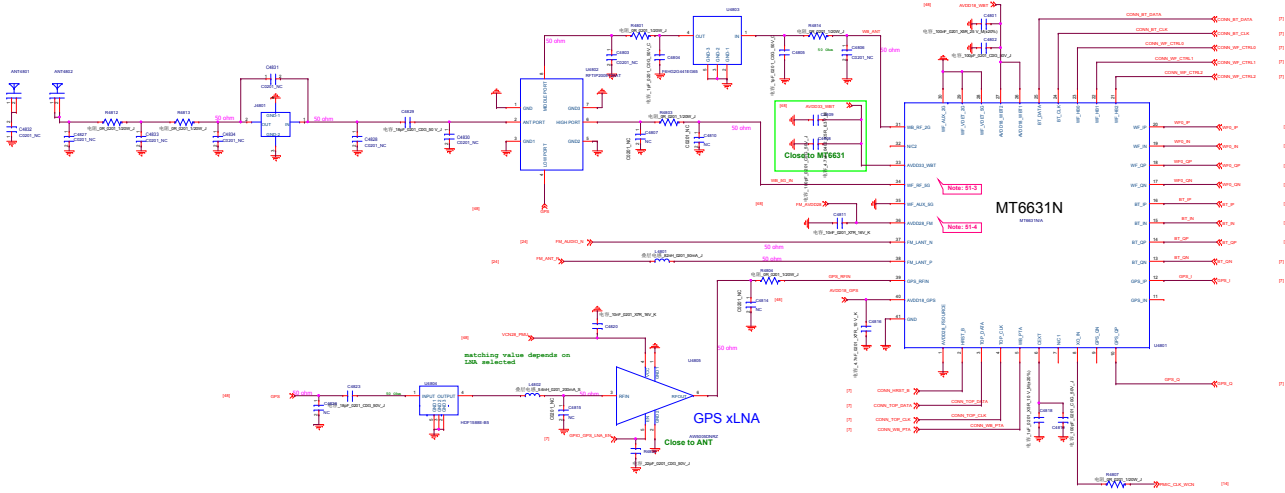
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ANT7
WiFi 2.4G & 5G_GPS

MT6631



Schematic design notice of "S1_CONNECTIVITY_CONSYS_MT6631"

Note 51-1: For R5015 size, please select 0402 size or larger one

Note 51-2: Please refer to MT6765 baseband design notice for VCN33_LDO selection guide

Note 51-3: If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND

Note 51-4: Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support